

NAME: A.K. LIKITH CHARAN
REG NO: 192412305
COURSE NAME: ANTENNA AND WAVE PROPAGATION

EX NO: 9B

SIMULATION OF SPACE WAVE PROPAGATION CHARACTERISTICS USING MATLAB

DATE: 11/08/26

OBJECTIVES

- i) To simulate Space Wave Propagation characteristics using MATLAB and analyze the variation of received signal strength with distance.
 - ii) To determine the Radio Horizon Distance and Line-of-Sight (LOS) communication range for different antenna heights.
 - iii) To analyze Free Space Path Loss (FSPL) and evaluate the performance of Space Wave communication systems using MATLAB.
-

OBJECTIVE 1

SIMULATION OF SPACE WAVE PROPAGATION CHARACTERISTICS

PROCEDURE

1. Open MATLAB and create a new script file.
 2. Define the transmitter power and operating frequency.
 3. Define the communication distance.
 4. Calculate the received signal strength.
 5. Plot signal strength versus distance.
 6. Analyze the propagation characteristics.
-

FORMULA

Free Space Path Loss (FSPL)

$$\text{FSPL} = 32.44 + 20 \log_{10}(f) + 20 \log_{10}(d)$$

Where:

- **f** = Frequency (MHz)
- **d** = Distance (km)

Received Power

$$P_r = P_t - \text{FSPL}$$

RESULT

The Space Wave propagation characteristics were successfully simulated and analyzed.

OBJECTIVE 2

DETERMINE RADIO HORIZON DISTANCE AND LOS RANGE

PROCEDURE

1. Define the transmitter antenna height.
 2. Define the receiver antenna height.
 3. Calculate the radio horizon distance.
 4. Determine the Line-of-Sight (LOS) communication range.
 5. Analyze the effect of antenna height on coverage.
-

FORMULA

Radio Horizon Distance

$$d_h = 4.12\sqrt{h}$$

LOS Communication Range

$$d_{\text{LOS}} = 4.12(\sqrt{h_t} + \sqrt{h_r})$$

Where:

- h_t = Transmitter antenna height (m)
 - h_r = Receiver antenna height (m)
-

RESULT

The LOS communication range was successfully determined for the given antenna heights.

OBJECTIVE 3

ANALYZE FREE SPACE PATH LOSS AND COMMUNICATION PERFORMANCE

PROCEDURE

1. Calculate FSPL at different frequencies.
2. Determine the received signal power.

3. Compare propagation performance.
4. Plot path loss characteristics.
5. Analyze communication coverage.

FORMULA

Received Power

$$P_r = P_t - \text{FSPL}$$

Link Budget Equation

$$P_r = P_t + G_t + G_r - \text{FSPL}$$

Where:

- G_t = Transmitter Gain
- G_r = Receiver Gain

RESULT

The free-space path loss and received power were successfully analyzed using MATLAB.

OVERALL MATLAB PROGRAM

```
clc; clear; close all;
```

```
%% Objective 1 : Space Wave Propagation
```

```
f = 900;           % Frequency (MHz)
```

```
Pt = 30;           % Transmit Power (dBm)
```

```
d = 1:1:100;       % Distance (km)
```

```
% Free Space Path Loss
```

```
FSPL = 32.44 + 20*log10(f) + 20*log10(d);
```

```
% Received Power
```

```
Pr = Pt - FSPL;
```

```
%% Objective 2 : LOS and Radio Horizon
```

```
ht = 50;           % Transmitter Height (m)
```

```
hr = 25;           % Receiver Height (m)
```

```
% LOS Communication Range
```

```
d_LOS = 4.12*(sqrt(ht) + sqrt(hr));
```

```

%% Objective 3 : Link Budget Analysis

Gt = 10;           % Transmitter Gain (dB)

Gr = 8;           % Receiver Gain (dB)

% Received Power using Link Budget

Pr_link = Pt + Gt + Gr - FSPL;

%% Display Results

fprintf('\nSPACE WAVE PROPAGATION ANALYSIS\n');

fprintf('-----\n');

fprintf('Operating Frequency = %.0f MHz\n', f);

fprintf('LOS Distance = %.2f km\n', d_LOS);

fprintf('Received Power at 10 km = %.2f dBm\n', Pr(10));

%% Plot Results

figure('Name', ...
       'Space Wave Propagation Characteristics', ...
       'NumberTitle','off');

% Free Space Path Loss

subplot(2,2,1);

plot(d,FSPL,'LineWidth',2);

grid on;

xlabel('Distance (km)');

ylabel('FSPL (dB)');

title('Free Space Path Loss');

% Received Power

subplot(2,2,2);

plot(d,Pr,'LineWidth',2);

grid on;

xlabel('Distance (km)');

ylabel('Received Power (dBm)');

title('Received Power vs Distance');

% LOS Communication Range

subplot(2,2,3);

```

```

bar(d_LOS);

grid on;

title('LOS Communication Range');

ylabel('Distance (km)');

set(gca,'XTickLabel',{'LOS Range'});

% Link Budget Analysis

subplot(2,2,4);

plot(d,Pr_link,'LineWidth',2);

grid on;

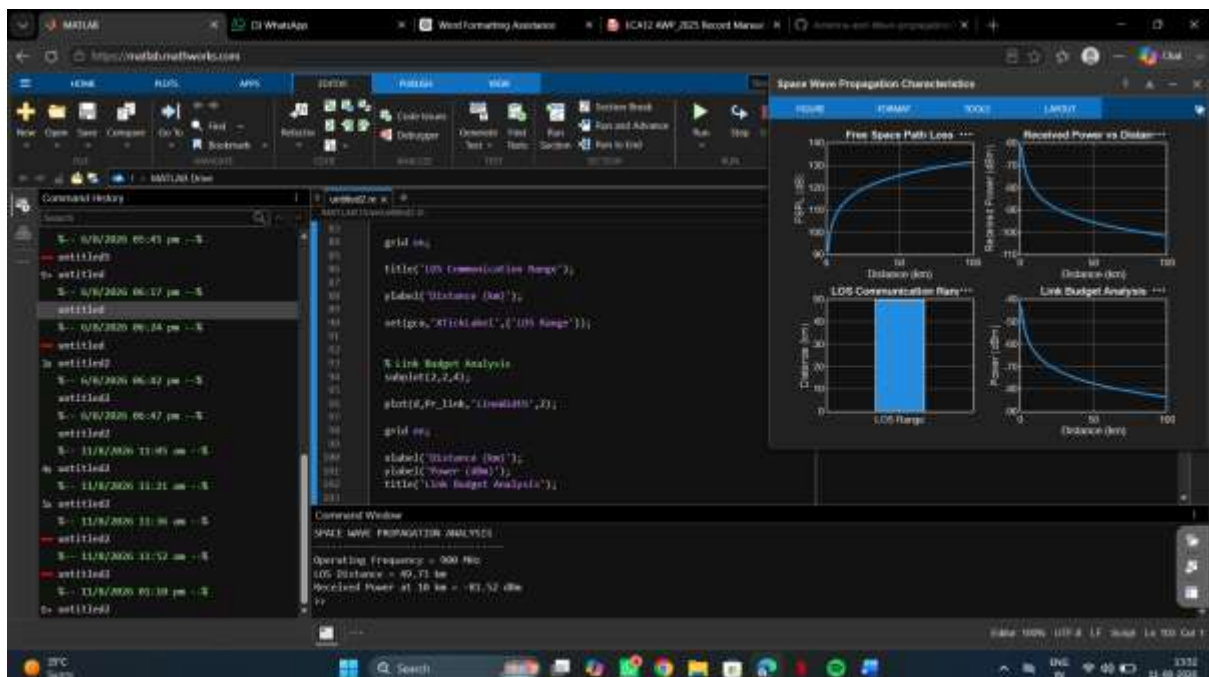
xlabel('Distance (km)');

ylabel('Power (dBm)');

title('Link Budget Analysis');

```

OUTPUT:



FINAL RESULT

The Space Wave Propagation characteristics were successfully simulated using MATLAB. The Free Space Path Loss and received power were analyzed as a function of distance. The Line-of-Sight (LOS) communication range was determined for the given antenna heights, and the link budget analysis was successfully performed.